Impossible or Just Unexpected? Testing What the Surprise of a Public V-JEPA Checkpoint Detects

Anand Rodriguez Menon
Department of Computer Science and Engineering
Amrita Vishwa Vidyapeetham, Amritapuri, India
am.sc.u4cse25025@am.students.amrita.edu

Dr. Anjali T
Department of Computer Science and Engineering
Amrita Vishwa Vidyapeetham, Amritapuri, India
anjalit@am.amrita.edu

Abstract—Violation-of-expectation (VoE) accuracy is often read as evidence that latent video world models represent intuitive physics. We ask what the underlying surprise signal of the public V-JEPA ViT-L checkpoint responds to, using three tests on frozen models: invariance to harmless edits, specificity against abrupt but possible events, and positive controls. On IntPhys dev, clean accuracy is 80.8% in fp16 (81.7% in fp32), well below the published 98% for the larger ViT-H model, for a reason we could not identify; a copy-last-frame pixel baseline reaches 76.1%. In fp16 inference, verdicts flip in 15.6% (strict ties) to 21.5% (tie-half) of pairs under an imperceptible one-grey-level dither. For near-threshold abrupt edits nominally matched in pixel magnitude, the surprise signal separates violations from edits with AUROC 0.546 (95% CI 0.501 to 0.591), below our pre-registered detection precondition, so we could not determine whether it tracks physical impossibility or unpredictability. Our positive controls were not met. On IntPhys 2 Main, descriptively, the checkpoint is at chance (47.6%). Results concern one public checkpoint, not latent world models in general.

Index Terms—intuitive physics, world models, V-JEPA, violation of expectation, video understanding

I. Introduction

The violation-of-expectation (VoE) paradigm shows a model two videos that differ in one respect. In a typical IntPhys scene a ball rolls behind a screen; in the possible video it comes out the other side, and in the impossible one it vanishes. The videos are otherwise matched pixel for pixel [1]. A model passes a pair if it is more surprised by the impossible video. The paradigm comes from developmental psychology and has been used to evaluate learned models of physics [2], [3].

For video models that predict in a learned representation space, surprise has a concrete definition: the model sees the first frames of a clip, predicts the representation of what comes next, and the surprise at each moment is the distance between that prediction and the representation of what actually happens. Garrido et al. applied this to V-JEPA [4] and reported a pairwise accuracy of 98% on the IntPhys dev set for their flagship ViT-H model, with the context length tuned per property, with pixel-prediction and multimodal language-model baselines near chance, and above-chance results on two further benchmarks (GRASP and InfLevel-lab), without physics-specific training [5]. This has been taken as evidence that predicting in latent space is enough to pick up some physical structure.

The accuracy number leaves a question open. A prediction error is large whenever the future is hard to predict. A ball that passes through a wall is impossible; a camera that jolts sideways is perfectly possible and still hard to anticipate. If the surprise signal responds to both, pairwise VoE accuracy on a benchmark whose impossible clips are also unpredictable could be high for reasons unrelated to physics. If it responds mainly to impossibility, the accuracy figure means what it appears to mean. Either answer is informative, and accuracy cannot distinguish them.

We train nothing. We test the surprise signal of frozen models directly, using deterministic edits to the IntPhys videos. The analysis plan and decision rules were pre-registered before any edited-condition output existed, with edit strengths calibrated on a separate split.

1) Invariance. Does the pairwise verdict survive edits that change appearance but not physics (blur, noise, compression, brightness, zoom, flips, frame dropping), applied identically to both clips of a pair?
2) Specificity. Does surprise separate real violations from physically possible abrupt events (a lighting step, a camera jolt, a sudden zoom) whose pixel change is scaled to match that of the violations, at near-threshold magnitude? Frame-difference baselines serve as a reference.
3) Positive controls. Does surprise rise when we create a violation ourselves, by flipping a clip vertically or reversing it in time? This checks that the pipeline can detect what it should.

We run these tests on the public V-JEPA ViT-L checkpoint, which is the confirmatory model, and descriptively on V-JEPA 2 [6], together with a randomly initialised network and trivial pixel baselines. Section III describes the protocol. Our findings are limits on what the instrument can show. The public checkpoint does not reach the published ViT-H accuracy, its verdicts are not stable under an imperceptible perturbation, our positive controls were not met, and the pre-registered specificity test could not reach the detection level it requires, so it

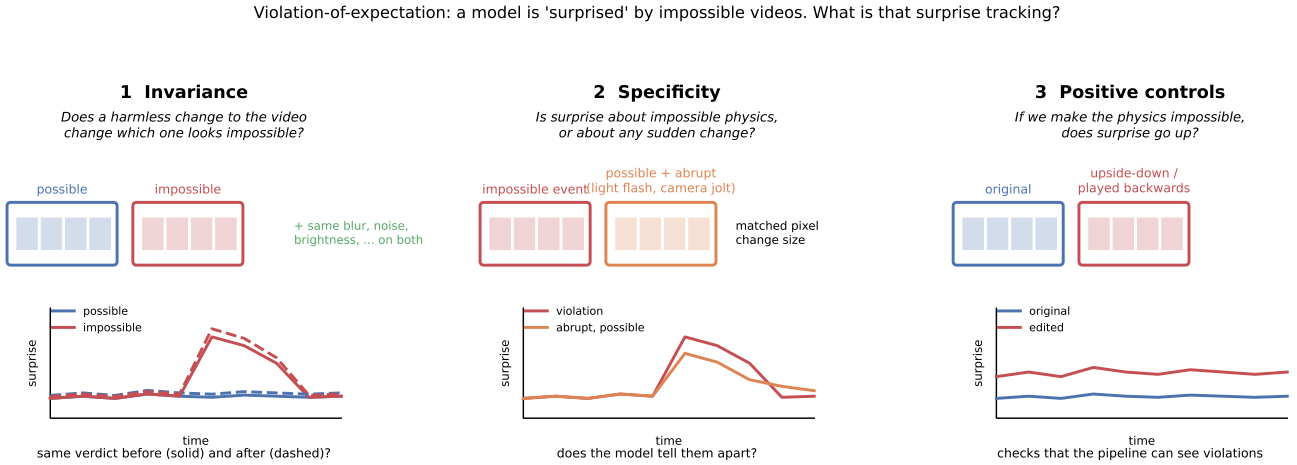


Fig. 1. The violation-of-expectation protocol and the three tests we run. A model watches a possible and an impossible clip that look almost identical, and its prediction error (surprise) is compared. We ask whether that comparison survives harmless edits (invariance), whether it separates physics violations from abrupt but possible events (specificity), and whether it rises when we create a violation ourselves (positive controls).

supports no claim in either direction. Sections IV and V report all of this, including what the pre-registration did not anticipate.

II. Related Work

VoE benchmarks. IntPhys pairs possible and impossible videos rendered in Unreal Engine, matched pixel for pixel, covering object permanence, shape constancy and continuity [1]. InfLevel provides VoE experiments for continuity, solidity and gravity, on which the video architectures it tested were at or near chance [3]. IntPhys 2 uses more complex scenes; V-JEPA and V-JEPA 2 are near chance on it (the best predictive model, V-JEPA 2, reaches 57.5%), and a qualitative debug scene shows no extra surprise at the point where an occluded ball should reappear [7]. Physion and Physion++ take another route: models predict object contact (and, in Physion++, infer latent properties such as mass and friction) and are compared with humans [8], [9]. We evaluate on IntPhys because its matched pairs make the pairwise verdict well defined.

Latent prediction and intuitive physics. V-JEPA learns video representations by predicting masked regions in representation space, with no pixel reconstruction [4]; V-JEPA 2 scales the recipe to more video and adds a robot-planning stage [6]. Garrido et al. used V-JEPA's prediction error as a surprise signal and reported 98% pairwise accuracy on IntPhys dev for their ViT-H model (context length tuned per property), with pixel-prediction and multimodal language-model baselines near chance [5]. VideoMAE V2, a masked autoencoder that reconstructs pixels, was planned as a pixel-reconstruction contrast model [10]; we did not run it. Joseph et al. probe video encoders layer by layer for physical information and find it concentrated at intermediate layers [11]. These works report that the signal scores well on one benchmark. We ask what it responds to, for a public checkpoint.

Doubts about what accuracy shows. Su et al. report near-chance IntPhys 2 accuracy (0.50–0.54) for V-JEPA models trained on three different video sources, and note qualitatively that frame-wise surprise was often similar for possible and impossible videos [12]. Li et al. find that on Physion, frozen V-JEPA 2 features with probes reach accuracy close to humans, while its disagreement with humans on individual items is far higher than between humans [13]. That is a contact-prediction task, not a VoE one. Both point to the same gap: accuracy alone does not say what the model uses. We add one control for the VoE setting, abrupt edits to possible clips whose pixel magnitude is matched to the real violations.

Counterfactual and likelihood-based evaluation of video models. CRONOS probes whether video generators stay physically consistent under controlled visual interventions such as viewpoint, scene and appearance, and measures generation quality, not model surprise [14]. Physics-IQ evaluates the physical understanding of generative video models [15]. LikePhys compares the denoising loss of matched valid and invalid videos for diffusion models [16]. YoCausal treats denoising loss as surprise and uses time reversal of real videos as a counterfactual for diffusion models [17]. We use time reversal as one positive control; those results do not tell us whether it works for latent predictors, so we test it (Section III). Our edits target the surprise of frozen predictive encoders.

Impossible versus improbable. Michaelov et al. found that language models do not reliably assign higher probability to possible-but-improbable sentences than to impossible ones [18]. Our specificity test asks the analogous question for video surprise. Video anomaly detectors likewise learn a model of normal footage, for example

normal driving, and flag deviations from it [19], with frame-level deep classifiers used to separate suspicious from normal activity in surveillance video [20]; whether such a signal separates a physical violation from any other unusual event is the question we ask of V-JEPA surprise.

III. Method

All models are frozen; we run inference only. Every claim below concerns the public V-JEPA ViT-L checkpoint unless stated otherwise.

A. Surprise

We reuse the surprise computation of Garrido et al. [5]. A predictor receives the representations of M context frames and predicts the representations of the next N frames. The surprise at time t is the L_1 distance between the predicted and observed representation, $s_t = \|\hat{z}_t - z_t\|_1$ (averaged over tokens). A clip's score aggregates the window surprises: the mean over windows for clean accuracy and invariance (H1), and the maximum for specificity and positive controls (H2, H3; event-aligned for H2). The default score takes, per window, the minimum over context lengths $C \in \{2, 4, 6, 8, 10\}$ frames. Ablations A1 (aggregation) and A2 (one fixed context length $C \in \{2, 6, 10\}$ at a time) are in Supplement Table S5. Models: V-JEPA ViT-L (M1) [5], [4], V-JEPA 2 ViT-L (M2) [6], VideoMAE V2 (M3, planned [10] but not run), M1 with random weights (M0r), and two baselines: frame-difference surprise $|f_t - f_{t-1}|$ (B1) and the pixel error of copying the last frame (B2).

B. Data

The main evaluation uses the IntPhys (2019) dev split [1], 180 pairs from 90 scene quadruplets (30 per block O1–O3). Edit strengths were calibrated on the 60-video IntPhys 2 Debug split [7], which is not used for evaluation. IntPhys 2 Main is used descriptively, for clean accuracy only; no edited runs were made on it (no event-time annotation). The IntPhys (2019) data are used as released for evaluation, and no videos are redistributed; the IntPhys 2 dataset card gives CC BY-NC 4.0. IntPhys 2 is used for evaluation only. In IntPhys dev, each impossible video is built by splicing frames of its possible partner videos (established by pixel hashing of the dev videos: 120 single and 60 double splices, none unspliced). This describes the dataset and is separate from the hypothesis, in the Discussion, that M1 relies on the splices.

C. Edits

Each edit is a deterministic function of frames, parameters and seed, applied before model preprocessing at three strengths (low, mid, high). Blur and low illumination at several graded severities are a common way to test vision models under degradation [21], [22]. Blur, JPEG and noise strengths are set on Debug frames to SSIM 0.95/0.85/0.70 (JPEG high, $q = 3$, reaches 0.689 because quality is an integer); brightness gain to a mean CIELAB L^* change of 5/10/20; centre-crop scale is fixed at 0.9/0.8/0.7. Horizontal flip, frame-rate halving and the dither floor have a single setting. Invariance edits are applied with identical parameters and seed to both videos of a pair: brightness/colour shift, Gaussian blur, JPEG re-compression, additive Gaussian noise, centre crop and resize, horizontal flip, frame-rate halving (degenerate, see item (f) of the deviations below). Abrupt edits are applied to possible videos only, at an event time t^*: a persistent lighting step, camera jolt, or sudden zoom. The event time t^* is the first frame at which the mean absolute pixel difference $D(t)$ between the impossible video and its prefix-sharing possible partner exceeds max(2/255, the 99.9th percentile of D over the first 10 frames of all If D never exceeds this floor, the rule falls back to a seeded draw from the middle half of the clip. Positive controls: vertical flip and time reversal of possible clips.

Magnitude matching. For each pair, the violation's pixel change is $|\Delta|_{\text{viol}} = |v_{\text{imp}} - v_{\text{pos}}|$ over frames after the event, and the edit strength is solved so that the edit's mean pixel change over the same window equals the pair's violation magnitude (closed form for lighting, bisection for jolt and zoom); mid strength is 1× this value, low 0.5×, high 2×. The pre-registered target is a median achieved-to-target ratio in $[0.95, 1.05]$ with at least 90% of pairs in $[0.9, 1.1]$. Matching uses dev pixels only, never model output.

D. Metrics and statistics

Pairwise accuracy is the fraction of pairs with $S(\text{impossible}) > S(\text{possible})$; flip rate is the fraction whose verdict differs from the clean one. We use McNemar tests against clean with Holm correction across edits. Specificity AUROC is $P(\Delta_{\text{viol}} > \Delta_{\text{abrupt}})$ with $\Delta_{\text{viol}} = S(\text{imp}) - S(\text{pos})$ and $\Delta_{\text{abrupt}} = S(\text{pos+edit}) - S(\text{pos})$, with a 10,000-resample bootstrap 95% CI; the D_1 and D_2 contrasts use a one-sided paired bootstrap p-value (floor 2×10^{-4}) and a paired CI. Detection rate for positive controls is the fraction of videos with $S(\text{edited}) > S(\text{original})$ with a binomial CI. Edit randomness uses 3 seeds (sign, colour direction, noise realisation, jolt direction); deterministic edits run once. The bootstrap resamples scene quadruplets, stratified by block. Ties are scored 0.5 ("tie-half") when two scores differ by at most 2×10^{-4} times the median clean score of that type; a strict convention ($<$, as in the released code) and an exact-tie convention ($\tau = 0$) are sensitivity analyses, and we report the tie-half and strict figures for the dither floor.

E. Pre-registration

Edit parameters, seeds, metrics, analysis plan and thresholds were frozen on 7 October 2026 at commit ee63aee (tag prereg-frozen), before any edited-condition output existed. Strengths were calibrated on IntPhys 2 Debug only. Decision rules, as frozen:

- Invariance (H1). An edit is "violated" if accuracy drops by at least 5 points with a Holm-adjusted scene-cluster bootstrap $p < 0.05$, or if its flip rate is at least 10% and at least 5 points above the dither floor with Holm-adjusted $p < 0.05$; it is "invariant" if the accuracy lower bound exceeds -5 points and the upper bound of flip minus floor is below 5 points. The Holm family has 14 tests (7 edits $\times$ accuracy and flip). If the dither floor is at least 10%, H1 for that model is descriptive.
- Specificity (H2). Let A be the specificity AUROC of M1, and A_0 the clean pairwise accuracy of M1 under the same event-aligned max aggregation (detection precondition). Define $D_1 = A_{\mathrm{M1}} - A_{\mathrm{B1}}$ and $D_2 = A_{\mathrm{M1}} - A_{\mathrm{B2}}$. "Specific" requires $A \geq 0.80$, both D_1 and D_2 significant under Holm correction over these two tests, and per-type AUROC of at least 0.65; $0.65 < A < 0.80$ is partial; $A \leq 0.65$ is not specific. If $A_0 < 0.85$ and the table yields not specific or partial, the outcome is "detection-limited": specificity cannot be assessed and no claim is made in either direction. This Holm-2 rule replaced an earlier intersection-union rule before the freeze.
- Positive controls (H3, gate G2). Vertical flip must raise M1's surprise in at least 80% of possible videos, with the lower confidence bound against horizontal flip above 0.5.

A model is H1-eligible only if its clean accuracy is at least 80.0%. Analyses outside this plan are labelled exploratory.

F. Deviations, amendments and known design weaknesses

Each item departs from the plan or limits what a result can show, and is carried into Section V. Full descriptions are in Supplement Section S-I. (a) Checkpoint gap (G1′). The original gate (at least 90% on IntPhys dev) was not met: the public ViT-L checkpoint reaches 81.7% (ViT-H 88.3%, tie-half under the pre-registered threshold). The checkpoints behind the published results are not released and the gap is unexplained; we do not claim a reproduction of the published 98%, which is for ViT-H with per-property context length. The replacement gate G1′ checks the pipeline only, and all three of its checks passed. (b) Positive controls (G2 not met). Vertical flip raised M1's surprise in 21.1% of possible videos against the 80% gate, and pairwise accuracy on flipped pairs is not lower than clean. Three readings remain open (the edits are weak violations for this model; the pipeline is insensitive; the signal acts as a splice-discontinuity detector), so M1's sensitivity to physical impossibility is not established. By human decision (Q10-A, logged 2026-10-08) the study continued with G2 not met, so every H1 and H2 statement carries this caveat. The random-weight network M0r responds in 87.8% of videos where about 50% was expected, so M1 minus M0r is not a measure of learned sensitivity. (c) Magnitude matching in H2. The frozen event rule fell back to a seeded random onset in 178 of 180 pairs. This is a failure of the rule, not a finding that no event exists. The median matched magnitude is 0.33 of 255 grey levels, so the edits are near-threshold; for global lighting, 84 of 180 entries are no-ops, so A is reported with and without that type. Both pixel baselines fall below 0.5 for a reason we did not verify. (d) M2 wrapper. The first V-JEPA 2 run used mask token index 1, but only token 0 is trained. After the freeze we changed it to 0 (human-approved, logged before any edited M2 output was opened) and reran the clean split: 46.9% before, 60.8% after. Both are reported; M2 is not H1-eligible. (e) Invariance is descriptive. The dither floor of 21.5% (15.6% strict) exceeds 10%, so every edit is read through its accuracy change and its flip rate relative to the floor. The floor was measured in fp16 only. For JPEG the verdict depends on the tie convention, and for zoom a retention rule leaves 112 of 180 pairs. (f) Frame-rate edit (design error). The pre-registered edit reproduces the skip-4 clean reference bit-identically, so its zero change cannot be read as invariance; it is excluded from all counts. (g) Other.

- Holm families. H1 primary: 14 tests. H1 dose-response: 30 tests for M1 (the pre-registration names 21, but only 15 graded cells exist) and 22 for M2 (11 cells, two tests each). H2 dose-response: 24 tests at cell level and 6 for the pooled contrasts. The M2 dose cells are seed 1 only.
- Precision. H1 eligibility was judged on the fp32 clean run and contrasts use fp16; the G1′ ladder compares ViT-L in fp32 with ViT-H in fp16.
- Logistics. Kernel bundles were rebuilt at later commits with the commit allowlist extended (human decision Q10-C: "proceeds"); the per-pair matching file was generated after the freeze from dev pixels only; the Colab fallback was not used. H2 was run as frozen, with its weaknesses disclosed, by human decision (Q11).

Exploratory analyses are labelled wherever they appear; the IntPhys 2 Main clean run is descriptive, as pre-registered.

IV. Results

Unless stated otherwise, results are for the public V-JEPA ViT-L checkpoint (M1) on the 180 IntPhys dev pairs. Percentages use the tie-half convention (Section III); strict-convention values are given where they differ. Confidence intervals are 95% scene-cluster bootstrap intervals. Under the pre-registered rules, H1 (invariance) is descriptive for M1, H2 (specificity) is detection-limited, and gate G2 was not met; read every statement below with that in mind.

A. Clean accuracy and baselines

Clean pairwise accuracy of M1 is 81.7% in fp32 (80.6% strict) and 80.8% in fp16 (81.1% strict); the fp16 run is the reference for all contrasts. The published 98%

TABLE I

Reproduction on IntPhys dev (180 pairs, 3 blocks): pairwise accuracy in % of the impossible video having the higher surprise (mean aggregation; tie-half = primary, strict = ties counted wrong). Published: Garrido et al. headline ≈98%, obtained with checkpoints that are not released; the gap to our public checkpoints is unexplained and we make no reproduction claim. M1 minus B2 (paired): +4.7 pp [−1.1, 10.6].

Model	Tie-half	Strict	Published
M1 V-JEPA ViT-L, fp32	81.7	80.6	≈98[a]
M1 V-JEPA ViT-L, fp16	80.8	81.1	–
M1H V-JEPA ViT-H (diagn.)[b]	88.3	87.8	–
M2 V-JEPA 2 ViT-L	60.8	60.6	–
M3 VideoMAE V2	not run		–
M0r random init (3)	50.0	50.2	expected ≈50
B1 frame difference	58.1	57.2	expected ≈50
B2 copy-last-frame	76.1	76.1	–
Superseded, disclosure only (not the M2 number)			
M2 old run (mask idx 1)[c]	46.9	46.7	–

[a]Literature value, not our run. [b]Prereg-tau tie-half 88.3; the 88.1 in the pre-registration text is the superseded fixed-τ figure (strict 87.8). [c]Wrapper deviation (query mask index 1), excluded from every M2 contrast.

(ViT-H) is not reproduced (Section III-F). The public ViT-H checkpoint, the like-for-like comparison, reaches 88.3% (87.8% strict). The copy-last-frame baseline B2 reaches 76.1%, and the windowed frame-difference baseline B1 reaches 58.1% (57.2% strict). The paired difference between M1 (fp16) and B2 is +4.7 pp with a confidence interval of −1.1 to 10.6 pp, which includes zero; against the fp32 run it is +5.6 pp (0.0 to 11.1), with the lower bound at zero. Clean accuracy does not separate M1 from a copy-last-frame pixel baseline. V-JEPA 2 (M2) reaches 60.8% after the mask-token correction (60.6% strict; 46.9% before it) and is not H1-eligible. VideoMAE V2 (M3) was not run. Table I lists all rows.

B. Invariance (H1): descriptive

The verdict flip rate under an imperceptible ±1 grey-level dither is 21.5% (15.6% strict), above the 10% limit, so invariance for M1 is descriptive and we report accuracy changes and flip rates relative to this floor (Table II; Fig. S1 in the supplement). We do not assign the driver's verdict labels. At high strength, additive noise lowers accuracy by −20.6 pp (−25.7 to −15.3) with a flip rate of 37.8%, +16.3 pp above the floor (10.6 to 22.4). Blur lowers accuracy by −14.4 pp (−19.4 to −9.4; flip 31.1%, +9.6 pp above the floor), and JPEG by −10.8 pp (−16.4 to −5.3; flip 28.3%, +6.9 pp above the floor). For JPEG the Holm-adjusted accuracy test depends on the convention: under tie-half the cluster and sign tests give p <0.0001 and 0.018, under the strict convention 0.0088 and 0.1151. Brightness changes accuracy by −2.4 pp (−5.1 to 0.3; flip 12.4%), horizontal flip by +2.2 pp (−2.2 to 6.7; flip 15.0%) and centre crop by −0.5 pp (−5.7 to 4.9; flip 14.3%, on 112 of 180 pairs after the retention exclusion). For these three, the flip rate is at or below the dither floor, so the edit adds no flips beyond the floor. The frame-rate edit is bit-identical to its reference by a design error and tests nothing; it is excluded here.

Dose-response (A3, descriptive). Accuracy loss grows with strength for blur (−4.7, −11.9, −14.4 pp at low, mid, high); for JPEG the change is +5.8, +0.8 and −10.8 pp, so a loss appears only at the highest setting. Noise is already at −17.7 pp at the lowest strength (flip 33.2%), and brightness stays within a few points of zero at all strengths, with low- and mid-strength flip rates (7.0% and 9.6%) below the floor. The family is Holm over 30 tests (Section III-F; Table S4 and Fig. S2 in the supplement).

C. Positive controls (H3, gate G2): not met

Vertical flip raises M1's surprise in 21.1% of possible videos (14.7 to 28.1), far below the 80% gate; against the horizontal-flip reference the rate is 25.8% (18.6 to 33.3). Horizontal flip itself gives 38.6%. Time reversal, reported but not gating, gives 60.8% (51.7 to 69.4). Pairwise accuracy on vertically flipped pairs is 83.9%, not lower than the clean 80.8%. The random-weight network M0r raises its surprise under vertical flip in 87.8% of videos (82.2 to 93.1), where the pre-registration expected about 50%; the M1 minus M0r difference in detection is −66.7 pp (−76.1 to −56.7), and we do not interpret it as a measure of learned sensitivity. G2 was not met. We cannot show that the pipeline responds to physical violations other than the dataset's own frame splices (Table S2 in the supplement).

D. Specificity (H2): detection-limited

The test uses the 180 possible videos. The event rule fell back to a random onset in 178 of 180 pairs, and the median matched magnitude is 0.33 grey levels of 255, so the edits are near-threshold. For global lighting, 84 of 180 entries are no-ops after rounding. The specificity AUROC of M1 is A =0.546 (0.501 to 0.591), and 0.520 (0.475 to 0.565) without global lighting. Per type it is 0.622 (global lighting), 0.480 (local lighting), 0.553 (camera jolt) and 0.528 (zoom); the global-lighting value largely measures clean detection and is not evidence of partial specificity. The detection precondition is not met: clean accuracy under the same aggregation is A_0 =0.722 (0.664 to 0.781) against the required 0.85. By the pre-registered rule the outcome is detection-limited: specificity cannot be assessed here and we make no claim in either direction. The outcome is the same under the strict and exact-tie conventions and with global lighting excluded. Ablations over aggregation and fixed context length are in Supplement Table S5, and results per block in Table S6.

Both pixel baselines fall below chance, with A_{B1} =0.335 and A_{B2} =0.379. The contrasts D_1 =+0.211 (0.159 to 0.263) and D_2 =+0.166 (0.116 to 0.219) are Holm-significant (p =0.0002, the bootstrap floor), but this reflects the baselines' distance below 0.5 under this matching, not evidence about M1; we do not read them as M1 outperforming a baseline. As a function of edit strength,

TABLE II

Invariance (H1), high strength. ΔAcc = accuracy under the same edit applied to both videos of a pair minus clean accuracy, in percentage points (scene-cluster 95% CI); flip = share of pairs whose verdict changes; flip$-f_0$ = flip rate minus the dither floor f_0 (a perturbation that should not matter at all): M1 f_0 = 21.5% (strict 15.6%), M2 f_0 = 24.6%. A flip rate near f_0 is what an invisible perturbation produces. Holm-14 adjusted p for ΔAcc / flip. M1 is the confirmatory row; M2 is descriptive only (clean accuracy below the 80% eligibility rule).

	M1 V-JEPA ViT-L				M2 V-JEPA 2 (descriptive)			
Edit (n pairs)	ΔAcc [CI]	Flip	Flip$-f_0$ [CI]	p (Δ/flip)	ΔAcc [CI]	Flip	Flip$-f_0$ [CI]	p (Δ/flip)
Brightness (180)	−2.4 [−5.1, 0.3]	12.4	−9.1 [−14.1, −4.4]	0.63 / 1.00	−0.4 [−3.1, 2.5]	15.4	−9.3 [−14.3, −4.6]	1.00 / 1.00
Noise (180)	−20.6 [−25.7, −15.3]	37.8	+16.3 [10.6, 22.4]	<0.001 / <0.001	−6.4 [−11.5, −1.3]	28.3	+3.7 [−0.2, 8.0]	0.13 / 0.30
Blur (180)	−14.4 [−19.4, −9.4]	31.1	+9.6 [4.3, 15.4]	<0.001 / 0.001	+12.8 [6.4, 19.2]	42.2	+17.6 [10.4, 24.8]	<0.001 / <0.001
JPEG (180)	−10.8 [−16.4, −5.3]	28.3	+6.9 [0.6, 13.7]	<0.001 / 0.15	−1.7 [−7.8, 4.4]	40.6	+15.9 [8.9, 23.3]	1.00 / <0.001
Zoom (crop)[a] (112)	−0.5 [−5.7, 4.9]	14.3	−8.9 [−16.4, −1.6]	1.00 / 1.00	+7.1 [1.7, 12.8]	25.0	−5.7 [−13.9, 3.1]	0.13 / 1.00
Horiz. flip (180)	+2.2 [−2.2, 6.7]	15.0	−6.5 [−12.8, −0.2]	1.00 / 1.00	+3.3 [−0.3, 6.9]	15.6	−9.1 [−14.1, −4.1]	0.58 / 1.00

Flip and flip$-f_0$ in %, ΔAcc in pp. [a]Pairs where the crop keeps the scene support (112 of 180); the other 68 are excluded for both models. Frame-rate halving is not tabulated: its edited input is bit-identical to the skip-4 clean reference, so the edit is degenerate by design (a design error in the edit list, reported here for completeness).

A is 0.594 (0.546 to 0.642) at half the matched magnitude, 0.546 at the matched magnitude and 0.501 (0.458 to 0.543) at twice, and it falls with strength in all four types (descriptive; Holm over 24 tests, all at the bootstrap floor). Figure 2 shows the distributions, and Table S3 in the supplement lists the per-type values, the M2 rows and the analysis without global lighting.

E. V-JEPA 2 (M2): descriptive

M2 is H1-ineligible (clean 60.8%), so all M2 results are descriptive. Its dither floor is 24.6%. Its specificity AUROC is 0.449 (0.403 to 0.496), 0.415 without global lighting, with A_0 =0.600 (0.542 to 0.658); the rule again gives the detection-limited outcome, and the upper interval bound lies below 0.5. Both pixel baselines are again below 0.5 (0.324 and 0.376).

M2 positive controls. Vertical flip raises M2's surprise in 71.9% of videos (64.4 to 78.9), reversal in 46.4% (38.6 to 54.2); G2 is not met for M2 either, and M2 minus M0r is −15.8 pp (−25.0 to −6.7). Pairwise accuracy on its vertically flipped pairs is 64.4%. The M2 dose-response (seed 1 only, descriptive) is in Supplement Section S-II. Because M2 is ineligible and does not meet G2, no statement here extends to latent world models in general.

F. IntPhys 2 Main: descriptive

On the 506 IntPhys 2 Main pairs, M1's clean accuracy is 47.6% (43.7 to 51.7; 47.8% strict), at chance. The pixel baselines are also at chance (B1 51.2%, B2 51.9%, B3 46.6%). By category M1 scores 46.7% (continuity), 42.1% (immutability), 49.6% (permanence) and 51.4% (solidity). Removing the four pairs whose names overlap the Debug split changes nothing material (47.6%). The verdict is at chance here, which limits what the signal can show; we draw no further conclusion from it. We did not run the specificity test on this split (no event-time annotation).

G. Exploratory analyses

These were run after the confirmatory results were seen and are exploratory (details and Table S1 in Supplement Section S-II). An onset-aligned specificity variant, using the splice time derived from prefix-sharing frames of the dev pixels, gives an AUROC for M1 of 0.392 (0.354 to 0.428), lower than with random windows (0.546), with the detection precondition unchanged (A_0 =0.722), so the rule's outcome does not change; both baselines stay below 0.5. An AUROC below 0.5 means that the matched possible edits raise surprise more than the violations do. M2's AUROC (0.449, upper bound 0.496) is also below 0.5. Under the frozen table this pattern carries the sub-label "Reversed"; because the detection precondition is unmet, the confirmatory M1 outcome stays detection-limited and we draw no conclusion from it. A hold-based frame-rate edit that differs from the skip-4 reference changes accuracy by +1.9 pp (−3.3 to 7.5) with a flip rate of 20.0%, within the 21.5% floor. The fp32 flip rate was not measured. A paired check on the stored arrays shows that M0r's surprise shifts systematically in one direction under vertical flip, opposite to M1's shift, and not through ties or noise; the cause is not established.

V. Discussion and Limitations

What the data show. For the public V-JEPA ViT-L checkpoint we can state four things. First, its clean pairwise accuracy on IntPhys dev (81.7%) is well below the published ViT-H figure (98%), and it is not separable from a copy-last-frame pixel baseline (76.1%; paired difference +4.7 pp, interval −1.1 to 10.6). Second, the verdict is unstable under a dither no viewer could see (21.5% of pairs flip, 15.6% strict), so any statement that a verdict is robust or fragile to an edit must be made against that floor. Third, our positive controls were not met: vertical flip raised surprise in 21.1% of videos against a gate of 80% (time reversal, not gating, 60.8%), while a randomly initialised network did respond to vertical flip (87.8%). Fourth, the pre-registered specificity rule returns "detection-limited": A =0.546 (0.501 to 0.591), and 0.520 (0.475 to 0.565) without global lighting, with a clean detection precondition of 0.722 against the required 0.85. Under the rule, specificity cannot be assessed here, and we make no claim in either direction: we do not claim that the

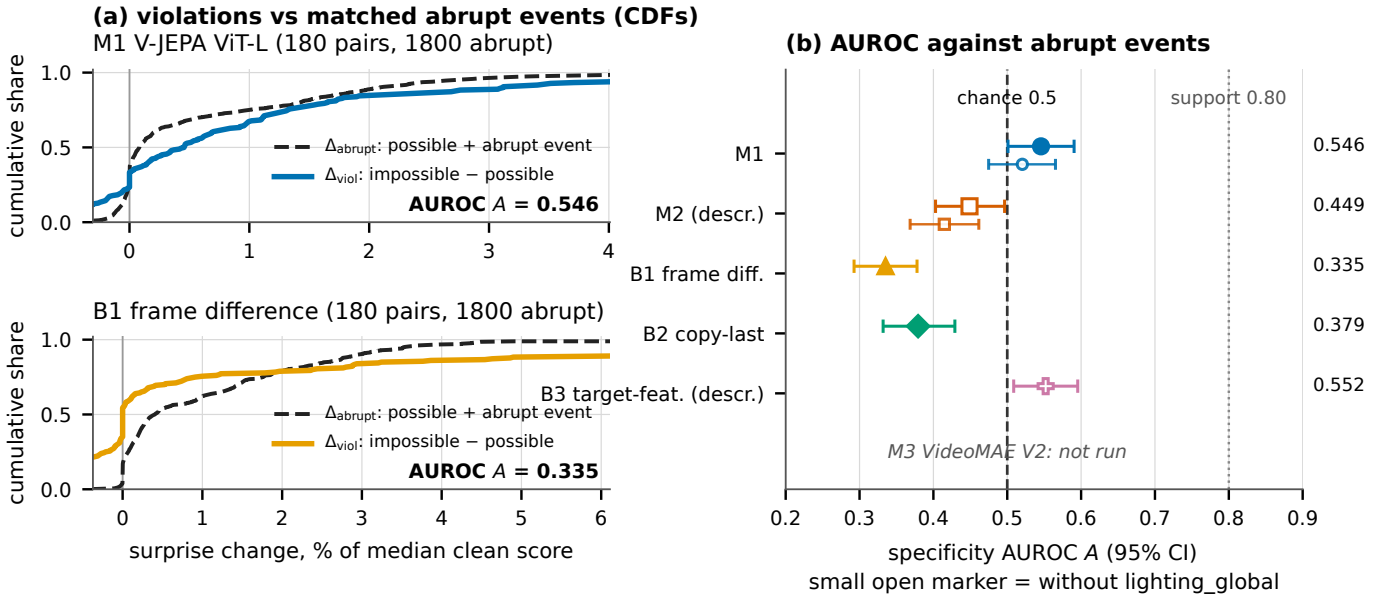


Fig. 2. Specificity (H2) for M1 and the pixel baselines: distributions of Δ_{viol} and Δ_{abrupt}, and AUROC with 95% intervals. Baselines below 0.5 reflect the matching, not model skill.

signal tracks unpredictability rather than impossibility, nor the reverse.

How these fit together. The results are consistent with a reading in which the pairwise verdict on IntPhys dev relies largely on the discontinuities that the dataset's frame splices create. That the dataset is built from splices is a measured property of the dev videos (Section III); that M1 relies on them is a hypothesis. A clip flipped vertically or reversed in time is internally consistent and has no such discontinuity. M1's surprise does not rise under vertical flip: in an exploratory paired check on the stored arrays its surprise decreases in 79.4% of possible videos, while the random-weight network's increases in 87.8%, and the cause is not established. Its accuracy on flipped pairs (83.9%) does not fall. Time reversal, which also contains no splice, raises M1's surprise in 60.8% of videos (lower confidence bound 51.7%); this modest rise is not what a splice-only reading predicts, so the hypothesis is at best partial. A copy-last-frame baseline, which also responds to discontinuities, reaches a similar accuracy; M1 and B2 are not distinguishable at 180 pairs, which is not evidence that they are equivalent. We label the splice reading a hypothesis: our data do not rule out other explanations, and we did not test it directly. The random-weight network's response indicates that vertical flip changes something an untrained network also responds to, so the flip control is weaker than intended whatever M1 does.

Exploratory splice analysis (post hoc, uncorrected, $n = 180$). Every one of the 180 impossible dev videos is a splice of its possible partners (120 single, 60 double). M1's accuracy is 91.2% on the 137 pairs that the copy-last-frame baseline B2 gets right and 47.7% on the 43 it gets wrong, and the per-pair margins of M1 and B2 correlate (Spearman 0.54, 0.42 to 0.65). M1 scores 75.0% on single and 92.5% on double splices, while B2 scores 75.0% and 78.3%. The window of peak surprise contains a switch frame in 73.9% of impossible videos against 46.9% expected by chance, and accuracy shows no monotone pattern across terciles of the splice position. Vertical flip changes few verdicts net (11 lost, 13 gained; McNemar $p = 0.84$), whereas time reversal loses 23 pairs and gains 9 ($p = 0.02$). These are associations. They do not separate "M1 detects the cut" from "M1 and B2 both respond to post-splice scene differences", and we do not take them as showing or establishing the splice reading. A causal test needs impossible videos that are not splices, which this dataset does not contain. On IntPhys 2 Main, M1 is at chance (47.6%, 43.7 to 51.7); this agrees in direction with earlier reports of near-chance V-JEPA accuracy on that benchmark [7], [12], and we draw no causal link.

What this implies for how VoE results are reported. A single pairwise accuracy cannot tell a model that tracks impossibility from one that tracks something correlated with it. Accuracy can also be inflated by how an evaluation set is built rather than by what the model has learned; slice-level data leakage in an MRI study is one documented example [23]. We suggest reporting, next to accuracy, (i) the noise floor of the verdict under an imperceptible perturbation, (ii) the accuracy of trivial pixel baselines on the same pairs, and (iii) a positive control, with a randomly initialised model alongside it. None of these needs training. Our matching of abrupt edits on mean pixel difference was not sufficient: both pixel baselines

landed below chance, so equal mean change does not mean equal visibility to every detector. A matched-event test with annotated event times and edits above the visibility threshold is a natural next step.

Scope. These conclusions concern one public checkpoint, on synthetic scenes, with frozen inference. V-JEPA 2 was run descriptively and is not H1-eligible, so nothing here extends to latent world models in general.

A. Limitations

These limit every conclusion; details are in Section III-F. (i) One public checkpoint, whose accuracy (81.7%) falls short of the published ViT-H figure for an unexplained reason. (ii) The positive controls were not met, so we cannot show a response to violations beyond the dataset's frame splices; in IntPhys dev a "violation" is a splice. (iii) The specificity edits are near-threshold (median 0.33 of 255 grey levels) and placed at random windows for 178 of 180 pairs, and global lighting is a no-op in 84 of 180 entries. (iv) The H2 tie threshold (1.09×10^{-4}) is small against the fp16 margin noise (up to 5.5×10^{-4} per pair), so many effects of this size lie near the numerical floor. (v) Both pixel baselines are below chance, so D_1 and D_2 are artefact-prone. (vi) Invariance is descriptive: the dither floor exceeds 10% and was measured in fp16 only, the zoom edit rests on 112 of 180 pairs, and the frame-rate edit is degenerate by our design error. (vii) M2 is H1-ineligible and descriptive, and M3 was not run. (viii) Seed variation covers nuisance parameters only and is not uncertainty; the scene bootstrap carries it. (ix) Synthetic scenes, one dataset for the main claims, frozen models, four abrupt-event types. We treat the models as frozen black boxes whose inputs we edit; attribution-style explainability analysis, which the authors have used for image-based sign-alphabet recognition [24], is left to future work.

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Reproducibility. The design was frozen at commit ee63aee; inference ran on Kaggle GPUs in fp16 with an fp32 check. Analysis code, the pre-registration and verification scripts are at github.com/anand-r-m/vjepa-surprise-physics; per-video surprise arrays are available on request. Videos are not redistributed.

References

[1] R. Riochet, M. Y. Castro, M. Bernard, A. Lerer, R. Fergus, V. Izard, and E. Dupoux, "IntPhys 2019: A benchmark for visual intuitive physics understanding," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 44, no. 9, pp. 5016–5025, 2022, arXiv:1803.07616.

[2] L. S. Piloto, A. Weinstein, P. Battaglia, and M. Botvinick, "Intuitive physics learning in a deep-learning model inspired by developmental psychology," Nature Human Behaviour, vol. 6, pp. 1257–1267, 2022.

[3] L. Weihs, A. Yuile, R. Baillargeon, C. Fisher, G. Marcus, R. Mottaghi, and A. Kembhavi, "Benchmarking progress to infant-level physical reasoning in AI," Transactions on Machine Learning Research, 2022.

[4] A. Bardes, Q. Garrido, J. Ponce, X. Chen, M. Rabbat, Y. LeCun, M. Assran, and N. Ballas, "Revisiting feature prediction for learning visual representations from video," arXiv preprint arXiv:2404.08471, 2024.

[5] Q. Garrido, N. Ballas, M. Assran, A. Bardes, L. Najman, M. Rabbat, E. Dupoux, and Y. LeCun, "Intuitive physics understanding emerges from self-supervised pretraining on natural videos," arXiv preprint arXiv:2502.11831, 2025.

[6] M. Assran, A. Bardes, D. Fan, Q. Garrido, R. Howes, M. Komeili, M. Muckley, A. Rizvi, C. Roberts, K. Sinha, A. Zholus, S. Arnaud, A. Gejji, A. Martin, F. R. Hogan, D. Dugas, P. Bojanowski, V. Khalidov, P. Labatut, F. Massa, M. Szafraniec, K. Krishnakumar, Y. Li, X. Ma, S. Chandar, F. Meier, Y. LeCun, M. Rabbat, and N. Ballas, "V-jepa 2: Self-supervised video models enable understanding, prediction and planning," arXiv preprint arXiv:2506.09985, 2025.

[7] F. Bordes, Q. Garrido, J. T. Kao, A. Williams, M. Rabbat, and E. Dupoux, "IntPhys 2: Benchmarking intuitive physics understanding in complex synthetic environments," arXiv preprint arXiv:2506.09849, 2025.

[8] D. M. Bear, E. Wang, D. Mrowca, F. J. Binder, H.-Y. Tung, R. T. Pramod, C. Holdaway, S. Tao, K. Smith, F.-Y. Sun, L. Fei-Fei, N. Kanwisher, J. B. Tenenbaum, D. L. K. Yamins, and J. E. Fan, "Physion: Evaluating physical prediction from vision in humans and machines," in Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2021, arXiv:2106.08261.

[9] H.-Y. Tung, M. Ding, Z. Chen, D. M. Bear, C. Gan, J. B. Tenenbaum, D. L. K. Yamins, J. E. Fan, and K. A. Smith, "Physion++: Evaluating physical scene understanding that requires online inference of different physical properties," in Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2023, arXiv:2306.15668.

[10] L. Wang, B. Huang, Z. Zhao, Z. Tong, Y. He, Y. Wang, Y. Wang, and Y. Qiao, "VideoMAE V2: Scaling video masked autoencoders with dual masking," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 14 549–14 560.

[11] S. Joseph, Q. Garrido, R. Balestriero, M. Kowal, T. Fel, S. Bakhtiari, B. Richards, and M. Rabbat, "Interpreting physics in video world models," arXiv preprint arXiv:2602.07050, 2026.

[12] E. Su, S. LeGris, T. M. Gureckis, and M. Ren, "Opinion: Learning intuitive physics may require more than visual data," arXiv preprint arXiv:2512.06232, 2025.

[13] F. Li, S. Zheng, Z. Yin, J. Cui, L. Ji, and J. Liu, "Human-level accuracy, non-human strategies: Revealing model–human divergence in video physical reasoning," arXiv preprint arXiv:2609.22788, 2026.

[14] L. Begiristain, O. Dünkel, and A. Kortylewski, "CRONOS: Benchmarking counterfactual physical consistency in video models," arXiv preprint arXiv:2605.23699, 2026.

[15] S. Motamed, L. Culp, K. Swersky, P. Jaini, and R. Geirhos, "Do generative video models understand physical principles?" arXiv preprint arXiv:2501.09038, 2025.

[16] J. Yuan, F. Pizzati, F. Pinto, L. Kunze, I. Laptev, P. Newman, P. Torr, and D. De Martini, "LikePhys: Evaluating intuitive physics understanding in video diffusion models via likelihood preference," arXiv preprint arXiv:2510.11512 (ICLR 2026), 2025.

[17] Y.-Z. Xie, Y.-H. Li, J.-Y. Lee, K. Zhang, Y.-L. Liu, and Z. Wang, "YoCausal: How far is video generation from world model? a causality perspective," arXiv preprint arXiv:2605.30346, 2026.

[18] J. A. Michaelov, R. Estacio, Z. Zhang, and B. K. Bergen, "Not quite Sherlock Holmes: Language model predictions do not reliably differentiate impossible from improbable events,"

in Findings of the Association for Computational Linguistics: ACL 2025, 2025, arXiv:2506.06808.

[19] S. Kandacharam, B. Rajathilagam, and S. K. Vasudevan, "Dual encoder-decoder-encoder with adversarial training for unsupervised traffic accident detection in surveillance videos," Journal of Visualized Experiments, no. 223, 2025.

[20] C. V. Amrutha, C. Jyotsna, and J. Amudha, "Deep learning approach for suspicious activity detection from surveillance video," in 2020 2nd International Conference on Innovative Mechanisms for Industry Applications (ICIMIA), 2020, pp. 335–339.

[21] K. S. Koushik, B. J. B. Nair, N. S. Rani, and M. Javed, "Robust classification of smartphone captured handwritten document images using deep learning," IEEE Access, vol. 13, pp. 14 922–14 948, 2025.

[22] C. Randieri, S. V. Ganesh, R. D. A. Raj, R. M. R. Yanamala, A. Pallakonda, and C. Napoli, "Aerial autonomy under adversity: Advances in obstacle and aircraft detection techniques for unmanned aerial vehicles," Drones, vol. 9, no. 8, p. 549, 2025.

[23] I. K. Veetil, D. E. Chowdary, P. N. Chowdary, V. Sowmya, and E. A. Gopalakrishnan, "An analysis of data leakage and generalizability in MRI based classification of Parkinson's disease using explainable 2D convolutional neural networks," Digital Signal Processing, vol. 147, p. 104407, 2024.

[24] V. Jayaprasad, A. R. Menon, and T. Anjali, "Towards inclusive AI: EfficientNet-based recognition of sign language alphabets," in 2025 5th International Conference on Evolutionary Computing and Mobile Sustainable Networks (ICECMSN). IEEE, 2025, pp. 220–225.